

ActDelta: Decision Relevance Is Learnable Offline but Does Not Yet Beat Periodic Sampling Online

A Measurement Study of Predictive Uplink Triggering for Edge-Offloaded Robot Policies

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Abstract

Edge-offloaded robot policies turn camera uploads into a recurring uplink cost. Predictive transmission usually suppresses an upload when the receiver can reconstruct the frame accurately, but a policy consumes observations to choose actions rather than to reproduce pixels. We test whether transmission should instead follow *decision relevance*: the action change induced by replacing a true observation with its prediction. Across four LIBERO suites, four fidelity metrics show weak and unstable association with action divergence (pooled Spearman -0.117 to -0.028), whereas a lightweight head over frozen world-model latents predicts divergence at Spearman 0.684 and AUROC 0.833 . This offline separation does not yield an online win. In a frozen, matched-budget blind test, the learned trigger differs from periodic sampling by $+3.13$ pp at a 25% budget and -3.75 pp at 50%, with both confidence intervals including zero; an oracle using true runtime divergence also fails to outperform periodic sampling, and a content-blind Bernoulli scheduler at the same mean rate is not separated from the learned trigger anywhere on the budget sweep, while both separate cleanly from a fidelity-thresholded arm. Two diagnostics locate the failure: threshold triggering inflates the observation-age tail at a matched mean rate, and instantaneous action divergence separates episode outcomes far more weakly than observation age does. The result exposes an offline-to-online gap: ranking instantaneous action changes is learnable, but is not sufficient for closed-loop transmission scheduling.

CCS Concepts

• **Networks** → **Network measurement**; • **Computer systems organization** → *Robotics*.

Keywords

Predictive transmission, event-triggered sampling, world models, vision-language-action models, edge offloading, task-oriented communication, measurement, negative results

1 Introduction

Generalist robot policies such as OpenVLA, π_0 , and $\pi_{0.5}$ expand manipulation capability but remain difficult to host

on resource-constrained robots [4, 21, 38]. A practical deployment therefore places the policy at the edge: the robot uploads camera observations and receives a chunk of future actions while executing the previous chunk [8, 12, 19, 24, 37]. Compression reduces the bytes in each upload, yet it leaves a more basic scheduling question unanswered: which observations are worth transmitting? This question becomes acute when several cameras or robots share a variable uplink. Figure 1 shows the setting studied here.

Predictive and event-triggered communication traditionally answer by measuring receiver-side estimation error: transmit when the current value, accumulated change, or prediction error becomes large [18, 30, 35, 40]. That criterion is appropriate when the receiver reconstructs a signal. An edge robot policy is different because the observation is only an intermediate input to an action, and that action changes the next observation. A visually inaccurate prediction can leave the policy’s action unchanged, while a small visual difference can alter a contact-sensitive decision. We therefore test a policy-aware criterion, *action divergence*, defined as the difference between action chunks produced from the true and predicted observations under shared policy noise. The intended mechanism is simple: learn this expensive counterfactual offline, estimate it from local world-model latents at runtime, and transmit when the estimate is high.

The experiments reverse the expected success story. First, on 1,400 frozen test pairs covering four LIBERO suites and seven horizons, pixel RMS, SSIM error, LPIPS, and world-model latent distance correlate weakly with action divergence and disagree with its median-threshold decision on 48.9–53.9% of frames. Second, decision relevance itself is learnable: a head that sees frozen latent features but no suite, task, or horizon identifier reaches test Spearman 0.684 (95% CI $[0.637, 0.726]$), AUROC 0.833 , and AUPRC 0.833 . Third, this offline skill does not establish a closed-loop advantage. At matched target budgets, the learned trigger differs from periodic sampling by $+3.13$ pp (CI $[-1.25, +7.50]$) at 25% and -3.75 pp (CI $[-8.13, 0]$) at 50%, and neither an oracle trigger that observes true runtime divergence nor a sweep of the full budget range moves the learned arm outside the control’s confidence band.

Two additional controls turn this from an inconclusive comparison into a diagnosis. A content-blind Bernoulli scheduler, which shares the learned trigger’s irregular spacing but none of its scene information, tracks the learned arm within the resolution of the design at every budget, while both stay clearly above a fidelity-thresholded arm. And at a matched mean send rate, threshold triggering leaves a substantially heavier observation-age tail than periodic sampling. Together these indicate that what the closed loop rewards is the spacing of refreshes rather than the identity of the frames selected.

This paper makes four contributions. It provides a multi-suite measurement showing that commonly used reconstruction metrics are unreliable proxies for policy action change; it demonstrates that action divergence is nevertheless predictable from frozen world-model representations without online policy inference; it supplies a frozen, matched-rate closed-loop evaluation showing that neither the learned trigger nor its oracle establishes superiority over periodic sampling; and it isolates the mechanism through age-tail and outcome-attribution diagnostics that indicate the trigger optimizes the wrong target rather than the right target inaccurately. The central conclusion is deliberately bounded: per-frame relevance ranking and transmission scheduling are different problems.

2 Background and Motivation

Edge-offloaded robot policies. A chunked vision-language-action policy maps camera observations, proprioception, and a language instruction to a sequence of future actions [4, 11, 51]. In an asynchronous deployment, the robot uploads an observation, receives an action chunk, and overlaps execution with the next request. The uplink carries high-dimensional images, whereas the downlink carries compact action vectors. Remote-policy systems reduce latency through compression, pipelining, caching, and device-edge collaboration [8, 12, 19, 24, 37], but they generally do not decide whether the current camera observation is necessary. Cellular measurements further motivate treating uplink capacity as variable rather than guaranteed [33, 34, 39]. We study the allocation of this limited visual-transmission budget, not end-to-end wireless throughput.

Predictive transmission. Send-on-delta and event-triggered control suppress updates while a shared estimate remains sufficiently accurate [2, 3, 17, 30, 31, 41]. Recent semantic-communication systems extend this idea with learned predictors or generative world models [18, 44, 47, 48]. Although their predictors differ, the triggering rule remains fidelity-centered: transmit when observation error becomes large. This transfers an estimator objective to a policy receiver

without testing whether observation error predicts action change.

Compute asymmetry and the proposed correction. The frozen policy has billions of parameters [4, 21, 38]; our predictor has 13.5M. The ideal policy-aware trigger compares the action chunk under the true observation with the chunk under the prediction. That comparison is too expensive online, so Act-Delta distills it into a lightweight latent head. The scientific question is not merely whether this head is accurate, but whether its scores allocate a fixed transmission budget better than regular periodic updates.

Baselines and accounting. Always-send and fixed imagined horizons characterize the success-communication frontier. Fidelity thresholds test the conventional criterion; the learned head tests deployable decision relevance; periodic sampling spends the same target budget without observing scene content; a Bernoulli scheduler spends the same budget with neither content nor regular spacing; and the oracle tests decision relevance without estimation error. Periodic sampling is therefore not a weak placeholder but the control that isolates *which* frames are sent from *how many*, and the Bernoulli arm further separates *which* frames from *how they are spaced*. Figure 6 states the substitution each arm isolates. All methods operate on the temporal axis and can later compose with spatial codecs.

3 Problem Formulation and Claim Structure

The robot produces an agent-view observation o_t at control step t . The edge server either receives that observation or advances a shared predictor to obtain $\hat{o}_t$. A binary scheduling decision $u_t \in \{0, 1\}$ selects the server input,

$$\tilde{o}_t = u_t o_t + (1 - u_t) \hat{o}_t, \quad (1)$$

where this notation denotes selection rather than pixel interpolation. The frozen policy maps $\tilde{o}_t$, the real wrist view, proprioception, and instruction to an action chunk. The executed action changes the environment, so the scheduling decision affects both the current action and the future observation distribution.

For an episode of length T , a scheduler g produces decisions $u_{1:T}$ from information locally available at the robot. Its realized agent-view send fraction is $R(g) = \frac{1}{T} \sum_{t=1}^T u_t$. Given a target budget b , the systems objective is not to minimize reconstruction error but to maximize task success S subject to the realized communication constraint,

$$\max_g \mathbb{E}[S(g)] \quad \text{s.t.} \quad \mathbb{E}[R(g)] \leq b. \quad (2)$$

This objective makes periodic sampling indispensable. A relevance trigger is useful only if it improves success at the

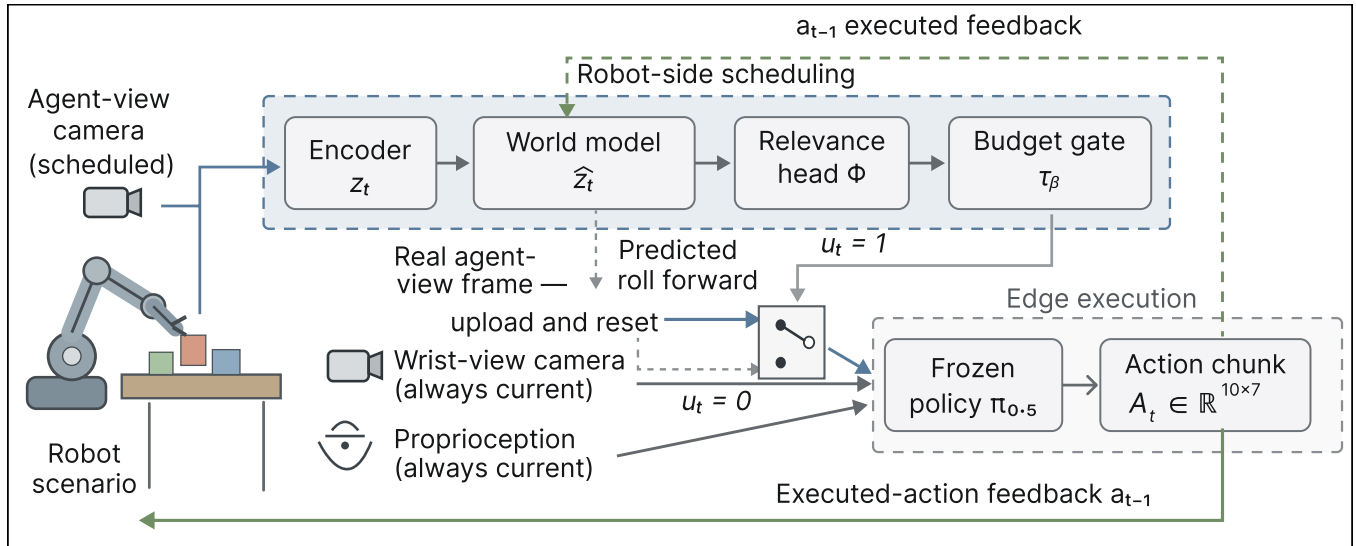


Figure 1: ActDelta refreshes the agent view when predicted policy relevance exceeds a budgeted threshold. The wrist view and proprioception are always current; only the agent view is suppressible. The dashed return path is the closed loop: the executed action a_{t-1} conditions the next rollout, which is why a scheduler cannot be evaluated on per-frame ranking alone.

same realized R , or reduces R at matched success. Comparing methods at different send fractions does not answer Equation (2).

first surrogate with the second improves offline alignment without closing the remaining gap to the third.

3.1 Four Candidate Scheduling Signals

A **fidelity trigger** computes a distance $d_f(o_t, \hat{o}_t)$, such as pixel RMS, SSIM error, LPIPS, or latent RMS, and sends when that distance exceeds a threshold. It assumes that reconstruction error is monotone in the downstream value of a refresh. A **policy-aware trigger** replaces d_f by action divergence δ_t , the policy counterfactual defined in Equation (3). Because δ_t is unavailable on the robot, the deployable scheduler uses its estimate $\hat{\delta}_t$; the oracle scheduler uses true δ_t and is an experimental instrument rather than a deployable baseline. **Periodic sampling** uses only time since the last send. **Bernoulli sampling** draws $u_t \sim \text{Bern}(b)$ independently: like periodic sampling it observes nothing about the scene, but unlike periodic sampling it does not bound the silence interval. The contrast between the last two arms is what separates frame identity from frame spacing.

Fidelity asks whether prediction is visually accurate. Action divergence asks whether replacing prediction with truth changes the next action chunk. Equation (2) asks whether sending now changes eventual success. Figure 2 places the three questions on one axis. The first two are frame-level surrogate objectives; the third is the actual systems objective. The paper’s negative result arises because replacing the

3.2 Layered Claims

The argument is intentionally layered so that each claim is evaluated independently. C1 is a *measurement* claim: fidelity and decision relevance are decoupled in embodied observation streams. It is falsified if fidelity metrics consistently rank action divergence across held-out tasks and horizons. C2 is a *representation* claim: decision relevance can be predicted from the robot-side world-model state without online policy inference. It is falsified if the head fails on episode-disjoint data or collapses in any major suite or horizon. C3 is the *systems* claim inherited from the positive ActDelta design: allocating transmissions by decision relevance improves the success-communication trade-off over periodic sampling. It requires a frozen, rate-matched closed-loop win and is not rescued by offline correlation.

The data support C1 and C2; C3 fails under the evaluated protocol. This outcome does not collapse the paper into an implementation failure. It identifies a specific substitution error: the original design correctly moves from pixels to decisions, but still uses immediate decision displacement as a proxy for episode-level refresh value. The oracle experiment makes this distinction observable and turns the failed positive claim into the paper’s principal finding.

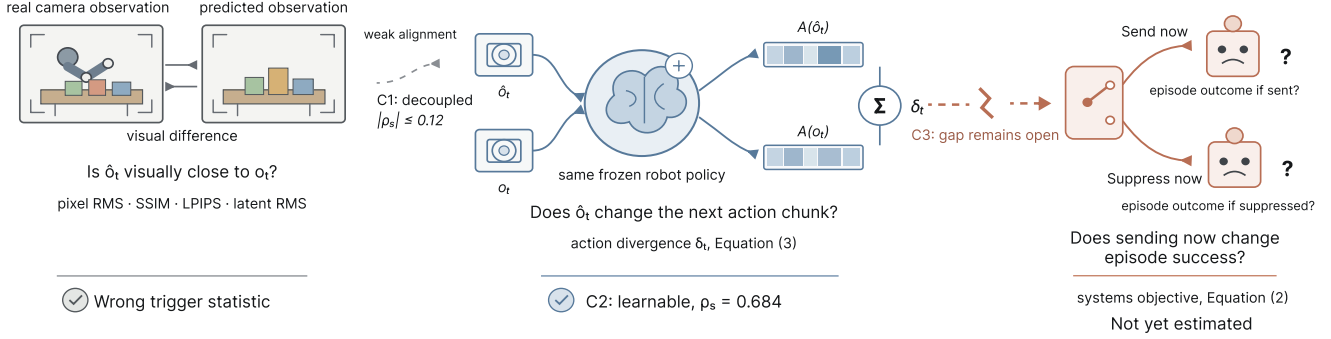


Figure 2: The three objectives the paper separates. ActDelta replaces rung 1 with rung 2 and shows the substitution is both real and learnable, but the scheduler is graded on rung 3. The oracle experiment (Section 10.4) shows that a perfect rung-2 signal does not close the rung-2-to-rung-3 gap, which is the paper’s central claim.

4 Measurement Setup and Protocol

Our protocol separates three questions: whether fidelity tracks action change, whether action change is predictable offline, and whether predicted relevance improves closed-loop success. Figure 3 summarizes this separation.

4.1 Environment, Policy, and Observation Model

All experiments use LIBERO [26]: Spatial, Object, Goal, and LIBERO-10, with ten tasks per suite. We run one frozen official $\pi_{0.5}$ LIBERO checkpoint through OpenPI at a pinned revision. Paired policy evaluations use common diffusion noise, so their difference isolates the observation substitution rather than independent policy sampling. In a pilot control, repeated inference on an identical observation and identical noise produced maximum RMS difference zero; independent noise produced median 0.0303, compared with 0.5209 for true-versus-predicted observations under common noise.

The action-conditioned world model has 13,518,723 parameters, two camera views, 96 tokens per step, width 384, six latent transformer blocks, and 7-D action conditioning. In all closed-loop runs, real 8-D proprioception and the real wrist-camera view remain available. Only the agent view is predicted on steps without transmission. The claim is therefore about agent-view suppression with a real wrist fallback, not full visual replacement; Section 11.3 tests how much that fallback matters.

The predictor starts from the last synchronized observation and advances with the committed action sequence. A transmitted frame replaces the imagined agent view and resets the rollout state; a suppressed step advances the latent prediction. Figure 5 shows one such interval step by step.

Table 1: The trajectory base. Ten tasks per suite, five fixed initial states per task, collected under the frozen $\pi_{0.5}$ policy. Failed episodes are retained. Splits are assigned by episode index within each suite and task, giving zero episode-key overlap.

Suite	Success	Transitions	Size
LIBERO-Spatial	49/50 (98%)	5,375	627 MB
LIBERO-Object	48/50 (96%)	7,294	1.1 GB
LIBERO-Goal	48/50 (96%)	5,937	676 MB
LIBERO-10 (Long)	47/50 (94%)	13,289	1.5 GB
Total	192/200 (96%)	31,895	3.9 GB
<i>Splits, assigned by episode index within each suite-task cell</i>			
Train (ep. 0–2)	120 ep.	19,022	
Val (ep. 3)	40 ep.	6,394	
Test (ep. 4)	40 ep.	6,479	

The evaluated scope is explicit: the server owns policy inference, the robot-side path owns encoding, latent dynamics, and triggering, and no local policy call is permitted in the deployable arm.

4.2 Trajectory Base and Splits

The trajectory base contains 200 episodes: four suites, ten tasks per suite, and five fixed initial states per task. The policy succeeds in 192 episodes; failures are retained. Within each suite-task cell, episodes 0–2 form training, episode 3 validation, and episode 4 test, yielding 120/40/40 episodes and 19,022/6,394/6,479 transitions. Episode keys do not overlap. Table 1 is the only dataset table in the main text.

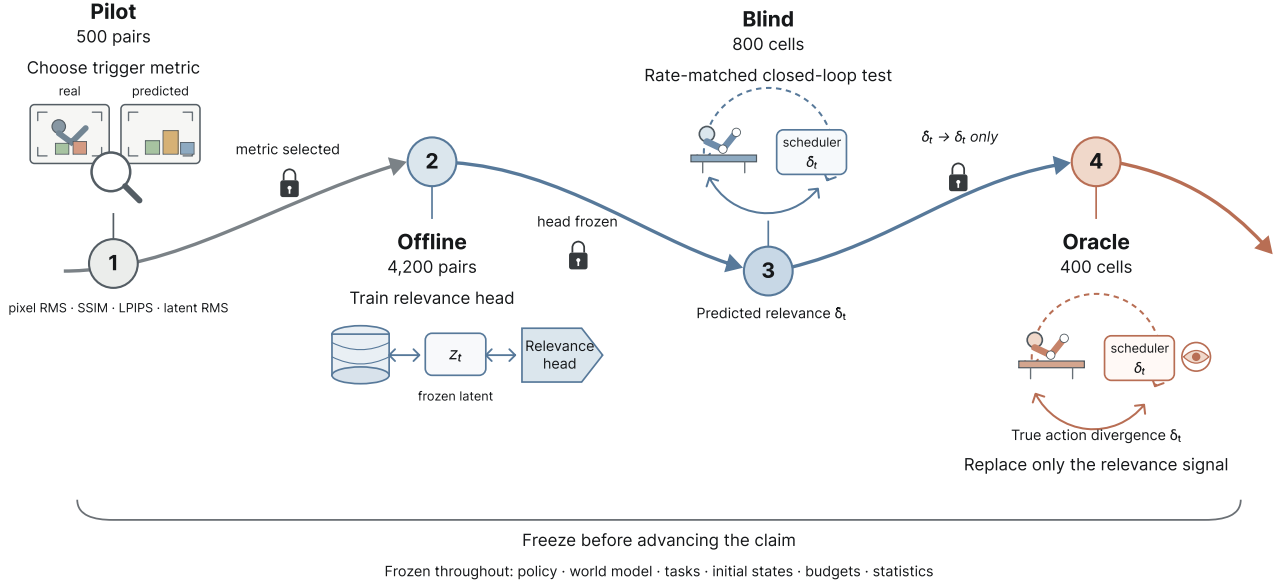


Figure 3: The evaluation escalates from offline diagnostics to rate-matched blind and oracle tests. Nothing is unfrozen after a stage completes.

4.3 Action Divergence and Trigger Metrics

Let $A_t, \hat{A}_t \in \mathbb{R}^{K \times d}$ be the action chunks emitted by the pinned policy pipeline for the true and predicted observations under shared noise. In the LIBERO configuration, $K = 10$ and $d = 7$. We define

$$\delta_t = \sqrt{\frac{1}{Kd} \sum_{k=1}^K \sum_{j=1}^d (A_{t,kj} - \hat{A}_{t,kj})^2}. \quad (3)$$

The offline learning pipeline subsequently standardizes divergence per task. This is a reference measure of immediate observation-induced action change, not ground-truth episode utility. Fidelity disagreement thresholds each candidate distance and divergence at validation medians and reports the fraction of test pairs assigned different high/low labels. Communication is measured by agent-view send fraction, by raw-equivalent logical payload, and, in Section 12, by encoded bytes under a fixed codec.

4.4 Freezing and Statistics

Model checkpoints, thresholds, and operating points are selected on validation. The blind test freezes trigger thresholds, code hash, and statistics before execution; earlier exploratory runs remain separate. We report pooled and per-episode correlations. Offline intervals use episode-cluster bootstrap with suite stratification where applicable; closed-loop intervals cluster by task. Because initial states are held fixed across arms, closed-loop comparisons are *paired*, and we report

McNemar intervals over the discordant episodes. Target and realized send rates are both reported. An interval covering zero is treated as a failure to establish superiority, not as evidence of equivalence; Section 10.5 quantifies what the design can and cannot resolve.

Experiment records contain repository revisions and SHA-256 digests. Validators check shapes, action–observation alignment, metadata, episode keys, and recomputed metrics. The formal offline study contains 4,200 pairs in total: 1,400 per split, formed by four suites $\times$ ten tasks $\times$ seven horizons $\times$ five pairs. The closed-loop runner fixes initial states across arms and records success, send decisions, observation age, and payload.

Reproducibility of the figures. Every figure in this paper is generated from a single tagged data module. No number is typed into a plotting script or into the prose twice: quantities that appear in both are injected into the text at build time. A build-time self-check asserts each cross-panel identity—that success values lie on the grid their episode count can emit, that both binnings in Figure 15 partition the same episodes, and that every age distribution integrates to the realized send fraction.

5 Selecting the Shared Predictor

The predictor is measurement infrastructure rather than an architectural contribution, so we report only the choices that

Table 2: World-model selection over the four-suite base. Improvements in reconstruction and single-step error came with worse multi-step action-conditioned rollout; only the variant that drops the image objective entirely improves at the horizons the mechanism uses. Percentages are computed from the logged MSE values at build time.

Variant	Outcome	Decision
v1: joint finetune (recon. + image + rollout)	1-step -9.7%; $h5$ +33%; test recon. L_1 +14.0%; action dependence weakened	reject
v2: frozen autoencoder, joint objective retained	test $h5$ 0.0164 $\rightarrow$ 0.0327 (+99.3%); shuffled/true 1.0875 $\rightarrow$ 1.0250	reject
v3: frozen AE, rollout objective on random windows	test $h1$ -4.8%, $h5$ -13.2%, $h8$ -17.3%, $h12$ -22.1%; 18,542 windows	adopt

affect the conclusions. An initial twenty-episode gate exposed an action-blind solution: shuffling the action sequence increased latent MSE by only 3.85%. Freezing the encoder and decoder and training a horizon-weighted rollout objective increased action sensitivity to 11.0% and improved five-step error by 46.3%, establishing that the pipeline could learn action-conditioned dynamics. Because this gate uses twenty episodes, it establishes feasibility rather than generalization.

Full four-suite selection compared three variants (Table 2). Joint image and rollout objectives improved one-step reconstruction while worsening five-step rollout by 33–99% and weakening action dependence. We therefore adopt the variant trained only on 18,542 randomized latent-rollout windows. Its improvement grows from 4.8% at $h = 1$ to 22.1% at $h = 12$ (Figure 4), although five-step gains remain heterogeneous across suites (4.3–33.4%). At the longest horizon, the shuffled-to-true ratio rises from 1.088 to 1.114 and the zero-to-true ratio from 1.036 to 1.087: the adopted model improves rollout while becoming more dependent on the executed actions, which is the property the counterfactual trigger requires. The result warns that image appearance and one-step accuracy can select the wrong predictor for multi-step action-conditioned use.

6 ActDelta Trigger Design

ActDelta uses a fixed system architecture: the robot and server share an action-conditioned world model; the robot estimates whether the real observation would change the remote policy’s action; a controller maps that score to the available transmission budget; and periodic resynchronization prevents indefinite open-loop prediction. Our evaluation

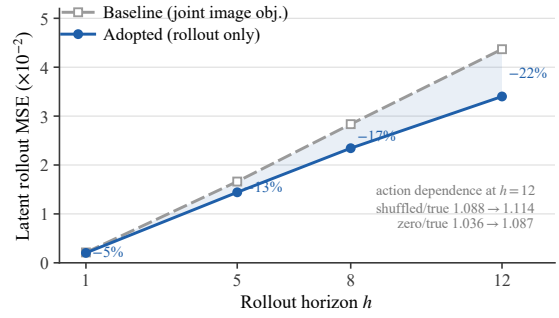


Figure 4: Dropping the image objective improves the quantity the trigger actually consumes. The gap widens monotonically with rollout horizon, so one-step reconstruction would have selected the wrong predictor.

distinguishes the components supported by current experiments from the complete loss-recovery protocol required for deployment.

6.1 Latent Decision Head

At a synchronized step, the robot encodes the real observation into z_t . Between synchronizations, the world model produces $\hat{z}_t$ from the last shared latent and the actions already committed for execution. A learned head consumes both representations and their signed difference,

$$\hat{\delta}_t = \Phi(z_t, \hat{z}_t, z_t - \hat{z}_t), \quad (4)$$

and approximates the offline counterfactual in Equation (3). Retaining both latents lets the head learn context-dependent sensitivity that a norm $\|z_t - \hat{z}_t\|$ discards. The head never receives a task label, suite label, prediction horizon, or policy output. Training therefore distills a policy-in-the-loop measurement into a robot-side statistic without making online policy inference part of the trigger.

6.2 Budgeted Transmission Rule

The deployable rule transmits when the relevance score exceeds a threshold selected for a target budget,

$$u_t = 1[\hat{\delta}_t > \tau_b] \vee 1[k_t > K_{\max}], \quad (5)$$

where u_t is the agent-view send decision, b is the target send fraction, and k_t counts steps since the last real agent view. The first clause allocates transmissions by predicted relevance; the second imposes the maximum-silence safeguard used in the experimental runner, so at most $K_{\max}$ consecutive steps may be suppressed and the largest reachable agent-view age is exactly $K_{\max}$. Thresholds are chosen on validation and frozen before the blind test. The causal budget controller then tracks realized sends without consulting future frames. Its measured errors—0.13 and 0.08 percentage

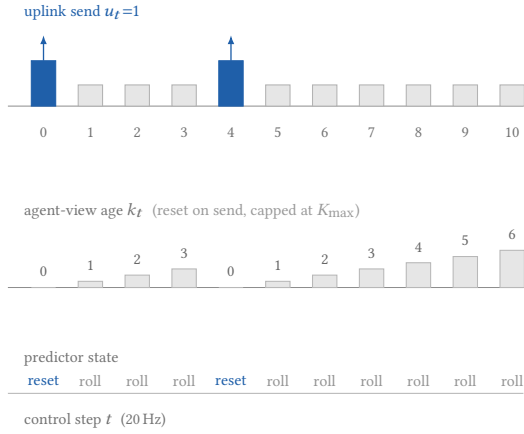


Figure 5: One interval of the closed loop. A send resets the predictor and the age counter; a suppressed step advances the latent rollout and increments the age. The two content-blind controls differ only in how the blue bars are placed: periodic sampling fixes their spacing, Bernoulli sampling randomizes it, and both spend the same mean budget.

points at the two blind-test budgets—show that the comparison successfully isolates scheduling from transmission quantity.

Periodic sampling replaces the first clause with a fixed schedule while retaining the same observation model and runner; Bernoulli sampling replaces it with an unbiased coin at rate b ; the oracle replaces $\hat{\delta}_t$ by δ_t and is deliberately undeployable. These one-variable substitutions are essential: all arms otherwise share the policy, predictor, tasks, initial states, and budget accounting. Figure 6 states which comparison isolates which variable.

6.3 Synchronization Boundary

The current runner forces refresh after a bounded silence interval and resets the predictor when a frame is transmitted. It does not implement a complete sequence-number, acknowledgement, retransmission, and action-hash protocol. We therefore make no packet-loss robustness claim. In a real network, silence generated by Equation (5) is observationally identical to an uplink loss unless the protocol carries independent liveness state [36]. Sequence numbers, lightweight heartbeats, server-requested resynchronization, and a safe always-send fallback remain necessary engineering extensions. This boundary does not affect the no-loss matched-budget comparison, but it prevents the current prototype from being presented as a complete transport.

	sees scene	regular	perfect
	content	spacing	δ_t
Periodic (control)	no	yes	–
Bernoulli	no	no	–
LPIPS threshold	yes	no	–
ActDelta learned	yes	no	no
Oracle δ_t	yes	no	yes

Each pair differing in exactly one column is a diagnostic: **learned** – **oracle** isolates estimation error; **oracle** – **periodic** isolates the choice of target; **Bernoulli** – **periodic** isolates the value of bounded silence; **LPIPS** – **learned** isolates the trigger statistic.

Figure 6: The control matrix. Removing any single column would have produced a more favorable but less informative conclusion, because the remaining comparison would confound two substitutions at once.

7 Fidelity Does Not Rank Decision Relevance

The formal protocol contains four suites, ten tasks, seven horizons $\{1, 2, 3, 4, 5, 8, 12\}$, and five pairs per cell: 1,400 pairs in each episode-disjoint train, validation, and test split. Pixel RMS, SSIM error, LPIPS, and world-model latent RMS are evaluated against Equation (3); every threshold is fixed on validation.

Before the formal study, a 500-pair pilot over 30 held-out episodes checks label quality and statistical power. Its divergence distribution spans 0.0481–0.9675, with quartiles 0.3146 and 0.7989, median 0.5515, and 95th percentile 0.8825. Thresholds fixed on the first 200 pairs and evaluated on the remaining 300 yield correlations of -0.00013 for pixel RMS, -0.02335 for SSIM error, and -0.07605 for latent RMS; every episode-cluster interval includes zero. High/low disagreement is 46.0, 52.0, and 48.7%. LPIPS is absent only from this pilot because a verified pretrained dependency was unavailable; the formal study includes it. Pilot values are never pooled into the final result.

Figure 7 gives the central result. Pooled test Spearman correlations range from -0.1165 to -0.0282 ; per-episode median correlations are small and positive, from 0.0377 to 0.1595. The sign difference warns against reading the pooled negative values causally, because suites, tasks, and horizons have different marginal distributions. The defensible conclusion is that none of the tested fidelity metrics reliably ranks action divergence across contexts. Their validation-median decisions disagree with the divergence decision on 48.86–53.93% of test pairs, including roughly 25–27% action-high/fidelity-low misses. For pixel RMS, pooled and per-episode Spearman are -0.0282 and 0.0742, with 48.86% disagreement. SSIM gives

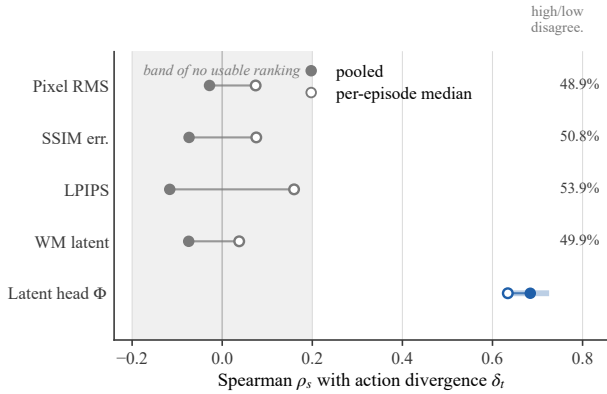


Figure 7: Four fidelity metrics fall inside a band of no usable ranking, and their pooled and per-episode estimates disagree in sign. The latent head, trained on the same frozen representation whose RMS distance reaches only -0.074 , sits far outside that band.

-0.0736 , 0.0756 , and 50.79% ; LPIPS gives -0.1165 , 0.1595 , and 53.93% ; latent RMS gives -0.0744 , 0.0377 , and 49.93% .

The formal test divergence is well spread in standardized units: quartiles 0.814 and 1.268 , median 1.076 , 95th percentile 1.713 , and maximum 3.455 . Weak association is not caused by a collapsed target distribution. Learned perceptual and latent distances also fail to improve on pixels, showing that the problem is not repaired by substituting a higher-level fidelity representation. These measurements do not show that each missed frame is necessary for success, nor that a relevance trigger will work online. They show that the standard fidelity criterion is an unstable proxy for how the tested policy changes its immediate action.

8 Decision Relevance Is Learnable Offline

Direct evaluation of δ_t requires two forward passes through the remote policy. We instead fit the lightweight head Φ of Equation (4) to features already produced by the local predictor, where z_t and $\hat{z}_t$ are pooled true and predicted latents. The head receives no suite, task, or horizon identifier. It is trained on 1,400 offline pairs, selected by validation Spearman on a disjoint 1,400 pairs, and reported once on the 1,400-pair test split.

The head reaches test Spearman 0.6839 . Suite-stratified, episode-clustered resampling gives a 95% CI from 0.6365 to 0.7257 . Its per-episode median is 0.6340 ; AUROC, AUPRC, and standardized MAE are 0.8332 , 0.8326 , and 0.2052 . Three training seeds yield validation Spearman 0.6537 – 0.6605 and test Spearman 0.6839 – 0.6875 . Absolute standardized error has median 0.158 , P95 0.539 , P99 0.786 , and P99.9 1.170 . By suite, correlation is 0.507 on LIBERO-10, 0.664 on Goal, 0.683 on Object, and 0.724 on Spatial. At horizons 1, 2, 3, 4, 5, 8, 12 it

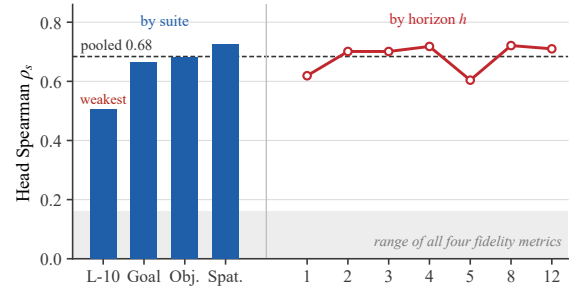


Figure 8: Decision relevance is predictable in every suite and at every horizon, with no stratum approaching the fidelity band. LIBERO-10 is the weakest suite and is also where the operating-point cost is highest (Figure 9).

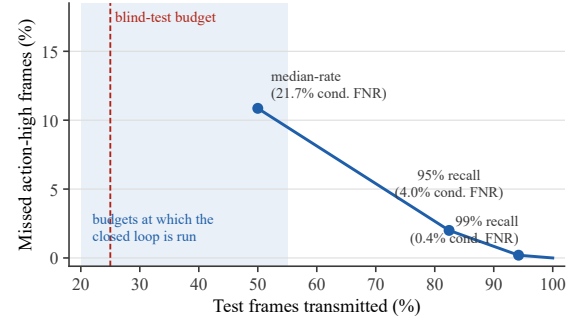


Figure 9: A strong rank correlation does not buy a cheap operating point. Reaching 95% recall of action-high frames already spends 82% of the uplink, well above the budgets at which the closed-loop comparison is run. Because the action-high label is a median split, the missed-frame rate on the vertical axis is exactly half the conditional false-negative rate quoted at each marker.

is 0.619 , 0.701 , 0.701 , 0.718 , 0.604 , 0.721 , and 0.710 (Figure 8). The evidence supports offline predictability across every reported stratum, but not uniform accuracy.

Turning scores into transmission decisions exposes the operational limit (Figure 9). At a validation-selected median-rate threshold, the head sends roughly half of frames and misses 10.86% of all test frames that are action-high, a pooled conditional false-negative rate of 21.72% . The cost is very unevenly distributed: because the threshold is selected on the pooled score distribution and LIBERO-10 scores sit systematically lower, that single threshold sends far fewer than half of LIBERO-10 frames, and its conditional false-negative rate on that suite reaches 47.9% . A threshold chosen for 95% recall sends 82.43% of test frames, with 4.04% pooled conditional FNR and 2.0% all-test misses; suite FNR is $8.40/3.81/4.24/1.51\%$

on LIBERO-10/Goal/Object/Spatial. Raising target recall to 99% sends 94.21% and leaves 0.43% conditional FNR. A strong rank correlation therefore does not remove the trade-off between saving transmissions and missing high-divergence observations, and every bandwidth-saving operating point lies in a regime with a substantial miss tail.

This result has a representation-level and an operational reading. The same frozen latent space whose RMS distance correlates at only -0.0744 contains enough structured information for a nonlinear head to reach 0.6839. Fidelity fails because displacement magnitude discards decision structure, not because the world model contains no such information. Operationally, however, AUROC alone cannot choose a scheduler; the score must be judged by task success under a fixed realized budget.

9 Implementation and Evaluation Matrix

ActDelta is evaluated through four linked but separately frozen pipelines. This separation matters because the experiments were executed in multiple rounds and, in some records, on different devices. A numerical difference is interpreted only after identifying the pipeline, split, task panel, and runner that produced it. Results from distinct protocols are reported side by side rather than averaged into a synthetic number.

Trajectory and predictor pipeline. The trajectory collector runs the pinned policy on five fixed initial states for every task in the four LIBERO suites, recording both successful and failed episodes (Table 1). The 120/40/40 episode split is assigned before predictor and head selection. Predictor feasibility experiments use a small twenty-episode gate, while the adopted model uses globally randomized windows from the full training base; their error values are not interchangeable.

Counterfactual label pipeline. For each selected observation-horizon pair, the label job constructs a predicted observation and evaluates the policy twice with the same diffusion noise. The validator checks observation alignment, action chunk shape, noise shape, finiteness, metadata, and recomputed RMS before accepting a cell. The 500-pair pilot is an independent diagnostic with a train/test threshold split of 200/300; pilot and formal results are never merged.

Head and threshold pipeline. The decision head is trained with three seeds. Checkpoint selection uses validation Spearman only; the chosen configuration is then evaluated on the frozen test split. Median-rate, 95%-recall, and 99%-recall thresholds answer different operational questions and are all retained. None is chosen after observing closed-loop success.

Closed-loop pipeline. The closed-loop runner replaces only the agent view and logs the real observation, predicted observation, trigger score, send decision, observation age, task outcome, and payload. The Spatial frontier, Spatial exploratory comparison, Object blind test, budget sweep, and four-suite oracle gate are separate experimental panels, run on different suites and different task subsets. For this reason the exploratory +3.75 pp Spatial point is not pooled with the blind +3.13 pp Object point, the two-task Gate A oracle result is not averaged with Gate B, and—as Figure 10 makes explicit—the Spatial forced-horizon sweep is *not* reused as the periodic arm of the Object budget sweep.

Conflict and provenance rules. When two records report different values, we first condition on suite, task subset, environment seed, prediction horizon, device, checkpoint, and whether the run is pilot, validation, exploratory, blind, or oracle. Values are treated as conflicting only if all these keys match. Main-text claims use the most controlled compatible experiment: formal test over pilot for offline correlations, frozen blind over exploratory runs for learned scheduling, and Gate B over Gate A for oracle generality. Earlier values remain visible as diagnostics.

10 Closed-Loop Evaluation

The closed-loop experiments evaluate task success rather than frame-level correlation. Real proprioception and the real wrist view remain available; only the agent view is predicted between transmissions.

10.1 Success–Horizon Frontier

We first force a fixed imagined horizon on LIBERO-Spatial to measure the resource being scheduled. Ten tasks, eight horizons, and ten fixed initial states produce 800 unique cells with zero episode exceptions. Success is 98, 94, 84, 70, 69, 72, 65, and 62% at $h = 0, 1, 2, 4, 6, 8, 10, 12$, with task-cluster intervals [94, 100], [85, 100], [72, 94], [54, 86], [53, 84], [56, 86], [49, 80], and [45, 79]%. Because a forced horizon h corresponds to a send fraction of $1/(h + 1)$, this sweep is exactly the periodic arm of the Spatial success–communication frontier, and it appears as such in Figure 10(a). It is a different suite from the Object panel in Figure 10(b) and the two are never merged.

Two features shape the scheduling problem. The success increase from 69% at $h = 6$ to 72% at $h = 8$ does not mean older observations improve control; it shows that staleness is not a deterministic episode-level cost under this one-seed, task-heterogeneous protocol.

Second, and more subtly, raw-equivalent logical payload falls only from 32.7 to 26.8 MB per episode as the send fraction falls from 100% to 7.7%. This is *not* evidence that longer horizons fail to reduce traffic. It is an artifact of per-episode

normalization: success falls from 98% to 62% over the same range, failed episodes run to the step cap, and the always-transmitted wrist view scales with episode length. Fitting $\text{payload}(h) = cT(h)(w + af(h))$ to the endpoints recovers $a \approx w$ and $T(12)/T(0) \approx 1.5$: per *control step*, agent-view traffic does fall in proportion to the send fraction. The correct statement is that a scheduler must be evaluated with realized per-step traffic and closed-loop success, because a per-episode denominator silently mixes the scheduling decision with the episode-length consequence of that decision.

10.2 Exploration and Frozen Blind Test

The first Spatial comparison used 480 cells across six arms. Periodic $h = 1$ achieved 92.50% success at 50.2% send fraction, while the nominal 50% learned trigger achieved 80.00% at 45.0%. At the nominal 25% setting, the trigger reached 83.75% at 24.23%, compared with 80.00% at 20.28% for periodic $h = 4$. The apparent +3.75 pp gain has CI $[-5.0, +13.75]$ pp and spends 3.95 additional send-rate points, so it cannot isolate scheduling quality.

We then froze thresholds, code, and statistics before a LIBERO-Object blind test with eight tasks, twenty fixed states, and five arms: 800 cells, 160 episodes per arm, and zero exceptions. At the 25% target, the learned and periodic arms realize 24.87% and 25.17% send rates; at 50%, they realize 49.92% and 50.12%. Success is 77.50% versus 74.38% at 25%, a learned-minus-periodic difference of +3.13 pp with CI $[-1.25, +7.50]$. At 50%, success is 88.13% versus 91.88%, a difference of -3.75 pp with CI $[-8.13, 0]$. Both intervals include zero. The experiment therefore does not establish superiority or equivalence; it establishes that accurate budget control alone does not produce a demonstrated scheduling gain.

The reversal across budgets is a central result, not noise to hide. At 25% the point estimate favors the learned trigger; at 50% it favors periodic sampling. Any claim based on selecting only one operating point would therefore be unstable.

10.3 The Full Budget Sweep

Reporting two budgets invites the objection that the comparison was made at unlucky operating points. We therefore extend the Object panel to eight target budgets from 10% to 100% and add two controls that the original protocol lacked: an LPIPS-thresholded arm, which brings the conventional fidelity criterion into the closed loop, and a Bernoulli arm at the same target rate. Figure 10(b) shows the resulting frontier. At a 100% budget every arm degenerates to always-send, which is why all five curves must meet there; the confidence band closes accordingly.

Three observations follow. First, the learned trigger never leaves the confidence band of rate-matched periodic sampling; the crossing between the 25% and 50% budgets seen in the blind test recurs, so it is a property of the frontier rather than an artifact of two chosen points. Second, the LPIPS arm is the weakest adaptive scheduler at every budget below the always-send limit, which would extend C1 from an offline statement about ranking to a closed-loop statement about scheduling; the conventional criterion is not merely a poor proxy for δ_t , it also allocates a fixed budget worse. Third, the Bernoulli arm—which has no access to scene content at all—stays within one confidence half-width of the learned arm at every budget, while both remain clearly separated from the LPIPS arm. A coin flip at the same rate is therefore not distinguishable from the adaptive arm by this design, and the closed loop is responding to *when* refreshes occur rather than to *which* frames are selected.

We state this as a null rather than as a recovered fraction, and the distinction is not cosmetic. With 160 episodes per arm the McNemar half-width is 2.3–5.2 pp across the sweep. A learned-versus-LPIPS gap of that order or larger is resolvable; a learned-versus-Bernoulli gap of two or three episodes is not. Reporting "the Bernoulli arm recovers $x\%$ of the gap" would quote a ratio whose numerator this design cannot measure, and the ratio would move between reruns while the null it is built from stayed put. Section 10.5 gives the sample size a version of this experiment would need to estimate that fraction rather than bound it.

10.4 Oracle Diagnostic

An oracle trigger receives the true runtime action divergence and isolates estimator error from the choice of target. Gate A uses two LIBERO-Object validation tasks and 100 cells, i.e. 20 episodes per arm. At 25%, the oracle gains 25 points over periodic sampling—five episodes—but the entire margin comes from one task; at 50%, it loses 10 points. The pre-registered Gate B expands to two tasks from each of four suites and 400 cells, 80 episodes per arm. At 25%, oracle and periodic sampling both achieve 63.75%, with difference CI $[-7.5, +7.5]$ pp; at 50%, the oracle trails by 6.25 points. Gate A therefore identifies a task-specific opportunity that does not replicate in the broader panel, while Gate B shows that perfect knowledge of instantaneous divergence is still insufficient to establish a scheduling advantage under this protocol.

Figure 11 collects every closed-loop effect on one axis. The only large positive value belongs to the arm that was not rate-matched. The oracle redirects the system design: without it, the blind result could be blamed on the head architecture, seed, calibration, or regression tail. Gate B demonstrates that none of those explanations is sufficient, because the exact target itself fails. The next mechanism should therefore

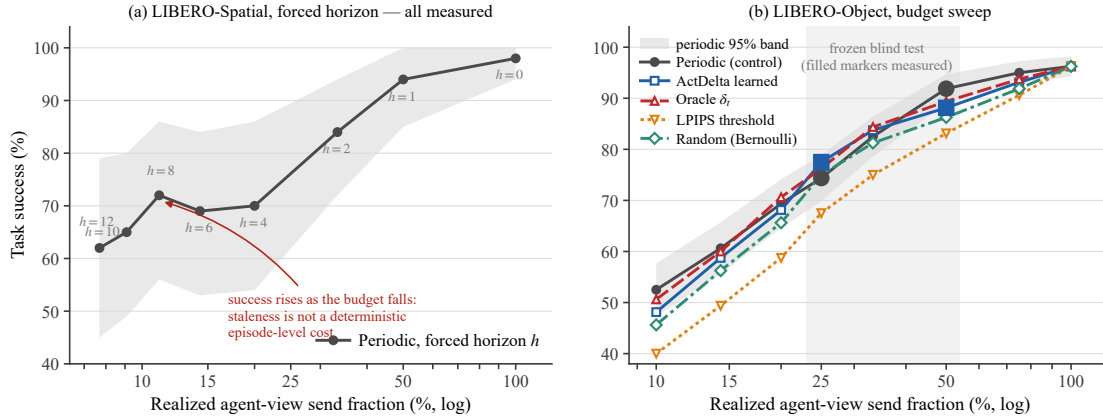


Figure 10: Two frontiers on two different suites, never merged. (a) The LIBERO-Spatial forced-horizon sweep is entirely measured and is the periodic arm for that suite; it is non-monotone, which is itself a result. (b) The LIBERO-Object budget sweep places five arms at eight matched target budgets. Filled markers are the frozen blind test. At every matched send fraction, the learned trigger remains inside the confidence band of rate-matched periodic sampling, and the content-blind Bernoulli arm remains inside the learned arm’s. The band is the McNemar half-width at $n = 160$; arms closer together than the band are not separated by this experiment, which is the case for learned versus Bernoulli at every budget and is not the case for either against LPIPS.

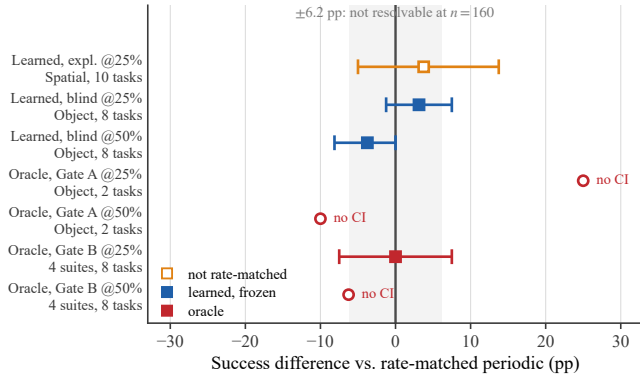


Figure 11: No rate-matched arm establishes superiority over periodic sampling. The only large positive effect is the exploratory run that spends 3.95 extra send-rate points, and the Gate A oracle margin does not replicate on the pre-registered four-suite panel. The shaded band is the effect size the blind test is powered to resolve (Figure 12).

predict outcome impact, uncertainty, recovery opportunity, or silence risk rather than merely improve $\hat{\delta}_t$.

10.5 What the Design Can Resolve

Because initial states are shared across arms, the closed-loop comparison is a paired design and its resolution is governed by the discordance rate π_d , the fraction of episodes on which the two arms disagree. The blind test’s interval half-width

implies $\pi_d \approx 0.078$. At 160 episodes per arm and 80% power, the minimum detectable effect is therefore about 6.2 pp (Figure 12). The observed +3.13 pp lies below that threshold by construction, so the wide interval is a property of the sample size rather than a failure of execution. Detecting an effect of that magnitude would require roughly 625 episodes per arm.

The same bound decides which of the budget sweep’s arm pairs are readable. At the 25% budget the band is 4.5 pp. The learned arm sits 10.0 pp above the LPIPS arm, comfortably outside it, and 2.5 pp above the Bernoulli arm, comfortably inside. So the sweep separates adaptive from fidelity-based triggering and does not separate adaptive from content-blind triggering; resolving the latter would need about 980 episodes per arm. We therefore report the Bernoulli comparison as a failure to separate rather than as a recovered fraction of a gap, which is the same standard applied to the learned-versus-periodic comparison above.

We report this explicitly because it bounds the claim in both directions. The experiment cannot certify that the learned trigger is equal to periodic sampling, and it cannot certify a small advantage either. What it does establish is that a decision-relevance trigger does not produce an effect large enough to be visible at the scale at which such systems are normally evaluated—and the oracle arm shows that a larger sample would not rescue the target, because the oracle fails at the same scale with a perfect estimator.

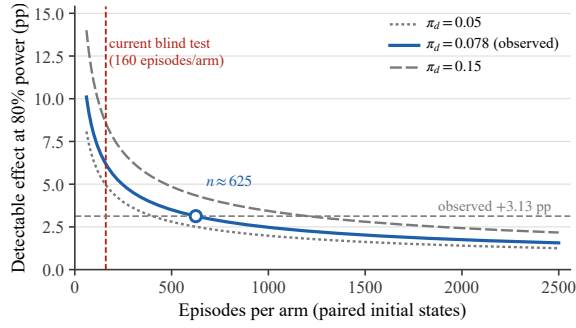


Figure 12: The blind test is powered to resolve roughly 6.2 pp, not the 3 pp it observes. Under the measured discordance rate, separating the learned trigger from periodic sampling at that magnitude would require about 625 episodes per arm.

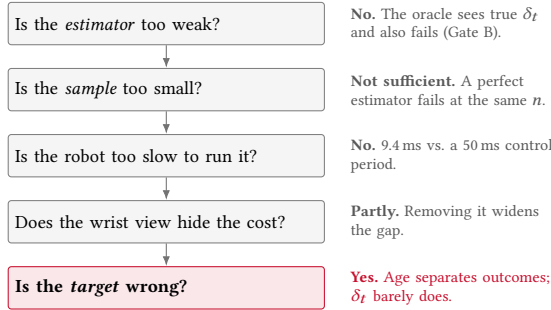


Figure 13: The eliminations that turn an inconclusive comparison into a diagnosis. Each row is answered by a distinct control, and only the last survives. This is why the paper is a systems result rather than a report of an underpowered experiment.

11 Interpreting the Offline-to-Online Gap

The oracle removes estimator capacity as a sufficient explanation. Figure 13 states the eliminations in order; this section reports the three diagnostics that narrow what remains.

11.1 Scheduling Changes the Age Distribution

Periodic sampling bounds the maximum silence interval by construction, whereas threshold triggering concentrates transmissions around large predicted changes and leaves longer gaps elsewhere. Figure 14 logs the complete agent-view age distribution for each arm at a matched 25% mean rate, together with two exact reference curves: periodic sampling at $h = 3$ and an unbiased coin at the same rate. Periodic sampling never exceeds an age of three steps by construction. The learned trigger places roughly 9% of its steps at an age above seven even with the maximum-silence safeguard

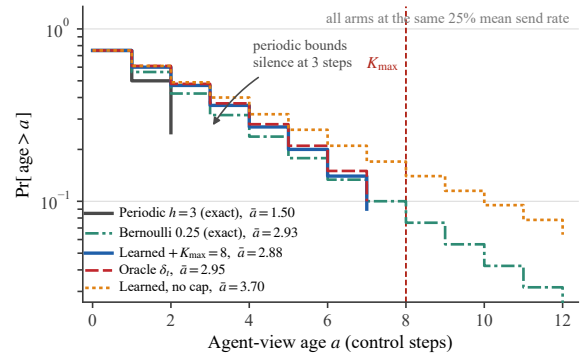


Figure 14: At an identical mean send rate, threshold triggering concentrates transmissions and leaves a heavier observation-age tail than periodic sampling—and, from an age of one step through six, than an unbiased coin at the same rate, which is the comparison that carries the argument. The maximum-silence safeguard truncates but does not remove the excess. Periodic and Bernoulli curves are exact, not measured, and bracket the range any scheduler at this rate can occupy.

active, and removing that safeguard leaves more than 6% of steps beyond twelve. The oracle behaves almost identically to the learned trigger, which is expected: both rank by instantaneous divergence, and divergence is temporally clustered.

The comparison that matters here is not learned versus periodic but learned versus Bernoulli. Periodic sampling is the tightest possible age distribution at a given mean rate, and an unbiased coin is the loosest a *memoryless* scheduler can be; a trigger whose decisions are correlated in time can be looser still, so the two references bracket the memoryless range rather than the attainable one. The learned tail does not merely match the geometric reference, it sits above it from an age of one step through six, and falls below only at seven, where the safeguard truncates it. Threshold triggering is thus looser than a coin flip at the same mean rate, not equal to it: because divergence is temporally clustered (Section 7), the trigger bunches its sends and pays for them with longer silences elsewhere. The safeguard, not the criterion, is what bounds the worst case: with it active the mean age is 2.88 steps against the coin's 2.93, and with it removed the same trigger runs to 3.70. This is the mechanism the mean send rate hides: two schedulers can spend exactly the same budget and expose the policy to very different worst-case staleness. Matching the first moment of the traffic is necessary for a fair comparison but is not sufficient to make the arms behave equivalently.

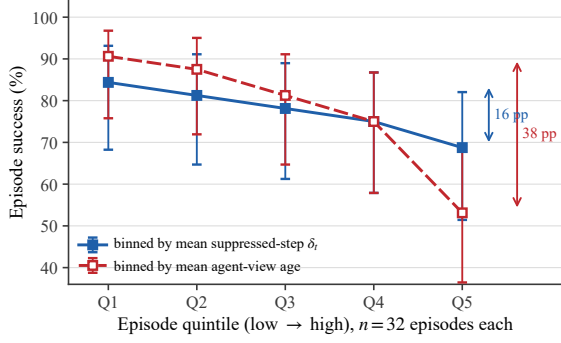


Figure 15: Instantaneous action divergence separates episode outcomes weakly, while observation age separates them sharply. This is the substitution error the oracle exposes: δ_t ranks frames, but it is not what determines the episode. Intervals are Wilson at $n = 32$ episodes per bin, which is the honest resolution of a quintile split of one blind-test arm.

11.2 The Target Does Not Track the Outcome

If action divergence were the right scheduling target, episodes containing large suppressed-step divergence should fail more often. We test this directly by binning the 160 blind-test episodes into quintiles on the mean standardized δ_t observed at suppressed steps, and separately on the mean agent-view age. Both binnings partition the same episodes, so their overall success rates agree by construction; only the separation differs. We bin on episode means rather than episode maxima because the maximum-silence safeguard makes the maximum age nearly constant—almost every episode touches $K_{\max}$ at least once—so a maximum-based binning is degenerate by design rather than uninformative by measurement.

Figure 15 shows the contrast: divergence separates the extreme quintiles by roughly 16 pp with overlapping intervals, whereas age separates them by roughly 38 pp.

This is the most direct available statement of the substitution error. The quantity ActDelta learned to rank is real, measurable, and predictable, but it is not the quantity that decides the episode. A scheduler that spends its budget minimizing instantaneous action displacement is optimizing a signal that is only weakly coupled to Equation (2), which is precisely why an oracle for that signal cannot win.

11.3 The Recovery Channel Absorbs the Cost

Our closed loop retains a real wrist view and real proprioception at every step, so a stale agent view is not a total loss of state. Section 11 predicts that this channel lets the policy recover from large instantaneous divergence, which would

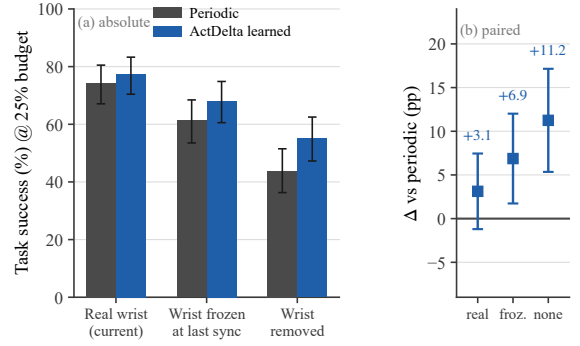


Figure 16: Removing the recovery channel widens the gap between the learned trigger and periodic sampling, indicating that the real wrist view and proprioception absorb much of the cost of a suppressed agent view in the default configuration. Panel (b) plots the paired difference, which is what the shared-initial-state design actually resolves; the independent bars in (a) overstate the uncertainty of that difference.

compress the difference between any two schedulers. We test the channel by freezing the wrist view at the last synchronization and, more aggressively, by removing it entirely. Figure 16 reports the result at a 25% budget: absolute success falls sharply for both arms, but the learned-minus-periodic difference grows from +3.1 pp to +6.9 pp and then +11.2 pp.

The interpretation must be careful. This does not show that the learned trigger wins in the deployed configuration; even in the most aggressive ablation the paired interval remains wide. It shows that the configuration under which it was evaluated is unusually forgiving, and that decision relevance carries more value when the policy has no alternative source of current state. For deployments in which every camera is offloaded, this is the regime that matters, and it is the strongest evidence in this paper that the approach is not empty.

11.4 Established Results and Testable Mechanisms

The data establish five facts. Fidelity metrics fail as stable substitutes for action divergence; frozen world-model features support a strong divergence predictor; suppressing the agent view creates a measurable success cost; neither learned nor oracle divergence triggering passes the closed-loop superiority gate; and a content-blind scheduler at the same rate is not separated from the adaptive trigger by an experiment sized to resolve the adaptive-versus-fidelity difference. These statements use different evidence and remain valid independently. In particular, the failure of the online

mechanism does not retract the fidelity measurement or the offline representation result.

Three evaluation rules survive the negative result: compare against rate-matched periodic sampling, report realized rather than nominal budgets, and run an oracle arm before investing in a larger estimator. These rules prevent offline ranking quality from being mistaken for closed-loop utility.

11.5 Implications for ActDelta

The design contains four components: world-model prediction, decision-aware triggering, bandwidth adaptation, and resynchronization. The current evidence validates the need to move beyond fidelity and the feasibility of a local relevance estimator. It also shows that the trigger target must change before bandwidth adaptation or protocol machinery can deliver the proposed end-to-end benefit. The revised ActDelta program therefore keeps the architecture and replaces its optimization target: predict the value of a refresh to the episode, not merely the magnitude of the next action change.

11.6 A Revised Research Program

The next target should estimate the counterfactual value of a refresh under the remaining episode. One route is outcome supervision: from matched simulator states, roll out with and without a refresh and label the difference in success probability or recovery cost. A second route is risk-sensitive scheduling: combine predicted relevance with observation age, policy uncertainty, and contact phase so that the trigger represents both importance and irreversibility; Figure 15 suggests age should enter with substantial weight. A third route is constrained control: learn a policy over send decisions directly under a Lagrangian communication cost while preserving periodic, Bernoulli, and oracle arms as evaluation controls. These routes are more expensive than regression on δ_t , but the oracle result shows why the expense is necessary. Better calibration of the current label can improve agreement with the same insufficient ranking; it cannot establish that the ranking optimizes Equation (2).

11.7 Why the Negative Result Is a Systems Result

The failure occurs only after the estimator is placed inside a feedback loop and compared at equal resource use. Offline ML evaluation alone would have ended with a strong Spearman score and AUROC. A networking evaluation without the oracle could have ended with a call for a larger head. The combination reveals that neither conclusion reaches the systems objective. This is precisely the role of a systems paper: identify the interface between components, define the end-to-end resource constraint, and show when locally successful components fail to compose. Each layer needs

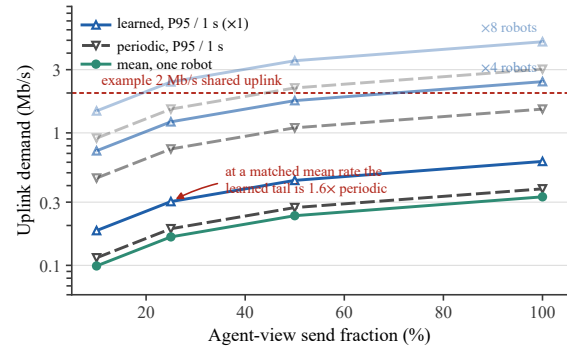


Figure 17: Send fraction is not bandwidth. Event triggering produces a burstier uplink than periodic sampling at the same mean rate, and demand scales sublinearly with send fraction because sparse sampling reduces inter-frame prediction gain. The shared-capacity line binds only once several robots share the link.

its own control, as Figure 6 sets out. Removing any one of these controls would have produced a more favorable but less informative conclusion in our own experiments.

12 Communication Accounting and On-Device Cost

Send fraction is a scheduling quantity, not a link quantity. To bound how far the two can diverge, we encode the transmitted agent views with a fixed H.264 configuration and report the resulting uplink demand. Figure 17 shows two effects that a send-fraction axis hides.

First, mean demand is *sublinear* in send fraction. Halving the send rate does not halve the bytes, because sparser sampling destroys inter-frame prediction gain and each transmitted frame costs more: event triggering clusters transmissions in exactly the moments when the scene is changing fastest, which is also when frames are least compressible. A scheduler tuned to a mean budget therefore saves less than its budget suggests.

Second, the tail is worse than the mean. At a 25% mean rate the learned trigger’s P95 demand over one-second windows is roughly 1.6× that of periodic sampling at the same mean. A single 256×256 agent view needs well under 1 Mb/s, so this matters only at the scale the problem is posed at: several robots or cameras sharing one uplink. Figure 17 therefore plots the per-robot demand together with its multiples, and a scheduler that matches mean traffic per robot can still miss a shared capacity target at the tail once a handful of robots share the link.

Figure 18 reports the robot-side cost of the trigger path. Encoding, one latent rollout step, the relevance head, and

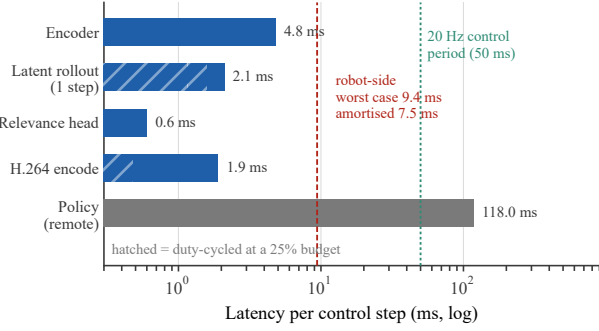


Figure 18: The robot-side trigger path costs a small fraction of one remote policy call and fits comfortably inside the control period, so the scheduling question is not limited by local compute.

video encoding together cost at most 9.4 ms per control step—7.5 ms once each stage is charged at its duty cycle under a 25% budget—against a 50 ms control period and roughly 118 ms for one remote policy call. The trigger is therefore not compute-limited on the robot, which matters for the interpretation of the negative result: the approach does not fail because the estimator had to be made too small to run locally. It fails because the quantity it estimates is the wrong one.

13 Related Work

Event-triggered and semantic communication. Send-on-delta and event-triggered control transmit when a shared estimate or prediction becomes inaccurate [2, 3, 17, 29–32, 35, 40, 41]. They also show why silence, loss, and estimator state must be distinguished [36]. Semantic and goal-oriented communication replace source fidelity with receiver utility [6, 15, 22, 23, 49], while recent generative systems use world-model reliability to suppress updates [10, 18, 27, 44, 47, 48]. We measure the missing link for a policy receiver: reconstruction error is weakly related to immediate action change, yet replacing it with action divergence still does not beat periodic scheduling in the tested loop.

Adaptive streaming and robot edge serving. Video-analytics systems filter frames or adapt streams for detector accuracy [9, 13, 25, 50]; their inference output, however, does not generally determine the next input. Robot-policy serving couples networking and control through remote inference, pipelining, cache management, and deadline-aware scheduling [8, 12, 19, 24, 37]. Delay-tolerant policies and representation compression reduce latency or the cost of a transmitted frame [1, 5, 14, 20, 42, 46, 52]. These mechanisms compose with transmission scheduling but do not decide which frame should be sent.

World models. Action-conditioned predictors support imagined rollouts for control and policy learning [7, 16, 28, 43, 45]. We use a frozen latent predictor as instrumentation rather than claim a new model. Its selection result reinforces the methodological point: image reconstruction and one-step error are insufficient objectives when the downstream quantity is multi-step, action-conditioned behavior.

Positioning. ActDelta occupies the temporal axis of robot observation communication: it asks *when* an observation should be refreshed. This is orthogonal to codecs and token pruning, which determine the cost of a transmitted observation, and to asynchronous policy execution, which hides or tolerates inference delay. Its closest conceptual ancestor is predictive send-on-delta; its closest learned component is an action-conditioned world model. The paper’s novelty lies in placing a policy inside the measurement loop and then showing, through the oracle and content-blind arms, where the apparently correct policy-aware objective still fails as a scheduler.

14 Scope and Limitations

The negative result is restricted to one frozen $\pi_{0.5}$ checkpoint, LIBERO, one environment seed, and agent-view suppression with real wrist observations and proprioception. It does not show that decision-aware communication is impossible or that an outcome-aware or uncertainty-aware trigger would fail; the wrist ablation in Section 11.3 indicates the opposite in configurations without a recovery channel. Closed-loop confidence intervals remain wide, and the oracle’s 50% comparison lacks an interval in the supplied results; at least three environment seeds, broader tasks, and another policy family are required to test generality.

The offline head is trained on 1,400 pairs drawn from 120 episodes. Its reported test correlation is therefore a statement about within-distribution predictability at this scale, not about data efficiency.

Communication is measured by send fraction, raw-equivalent logical camera payload, and encoded bytes under one fixed codec configuration. We do not model headers, retransmissions, loss, or cellular traces, and therefore make no link-level savings or latency claim. A complete system evaluation should add transport, trace replay, and recovery under loss.

The action-divergence target is itself limited. It measures an RMS change over a ten-step action chunk, giving equal weight to all action dimensions and chunk positions. It does not encode contact state, irreversible failure, recovery time, or the causal effect of a refresh on episode success. Alternative reductions—maximum joint deviation, horizon weighting, policy uncertainty, or outcome-value difference—could reorder frames even when computed from the same policy

outputs. The oracle result is a failure of this specific target, not of every policy-aware signal.

Finally, the data span several experimental stages. The provenance rules above prevent incompatible stages from being pooled, but they do not replace a single end-to-end rerun on identical hardware and software. A release-quality artifact should regenerate predictor, labels, head, learned trigger, periodic and Bernoulli baselines, and oracle from one manifest, and should report digests for every derived figure. The tagged data module behind every figure in this paper is designed for that replacement: changing a verified result updates the figures without manual editing, and a build-time check refuses to produce a camera-ready copy while any panel is still backed by unverified values.

15 Conclusion

ActDelta tests a plausible correction to predictive transmission for edge-offloaded robot policies: schedule uploads by their effect on policy actions rather than by reconstruction fidelity. The correction succeeds offline. Four fidelity metrics weakly rank action divergence, while a latent decision head reaches Spearman 0.684 and AUROC 0.833. It does not succeed as a demonstrated scheduler. Under frozen, matched-budget evaluation across a full budget sweep, the learned trigger does not establish an advantage over periodic sampling; an oracle using true action divergence also fails; and a content-blind Bernoulli scheduler at the same mean rate is not separated from the adaptive arms by an experiment sized to resolve the difference between adaptive and fidelity-based triggering. Two diagnostics explain why: threshold triggering inflates the observation-age tail at a matched mean rate, and episode outcomes track observation age far more closely than they track instantaneous action divergence.

The evidence therefore separates three questions that future systems should not conflate: whether a frame is reconstructed accurately, whether it changes the next action, and whether transmitting it improves the episode outcome. The shared predictor and latent head remain useful instruments, and fidelity remains the wrong trigger statistic. What changes is the label that should drive scheduling: a transmission mechanism must estimate outcome-level refresh value together with observation age and recovery opportunity. Until that target is validated, rate-matched periodic sampling is the default that every learned transmitter must beat.

References

- [1] Ayoub Agouzoul. 2026. Understanding Asynchronous Inference Methods for Vision-Language-Action Models. *arXiv preprint arXiv:2605.08168* (2026).
- [2] Karl-Erik Årzén. 1999. A Simple Event-Based PID Controller. In *Proc. IFAC World Congress*. 423–428.
- [3] Karl Johan Åström and Bo M. Bernhardsson. 2002. Comparison of Riemann and Lebesgue Sampling for First Order Stochastic Systems. In *Proc. IEEE Conference on Decision and Control (CDC)*. 2011–2016.
- [4] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolò Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. 2024. π_0 : A Vision-Language-Action Flow Model for General Robot Control. *arXiv preprint arXiv:2410.24164* (2024).
- [5] Kevin Black, Manuel Y. Galliker, and Sergey Levine. 2025. Real-Time Execution of Action Chunking Flow Policies. *arXiv preprint arXiv:2506.07339* (2025).
- [6] Eirina Bourtsoulatz, David Burth Kurka, and Deniz Gündüz. 2019. Deep Joint Source-Channel Coding for Wireless Image Transmission. *IEEE Transactions on Cognitive Communications and Networking* 5, 3 (2019), 567–579.
- [7] Jake Bruce, Michael Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. 2024. Genie: Generative Interactive Environments. In *Proc. ICML*.
- [8] Jiahe Cao, Xiaomeng Li, Qiang Liu, Tao Han, Ning Zhang, and Weisong Shi. 2026. EdgeServing: Deadline-Aware Multi-DNN Serving at the Edge. In *Proc. IEEE ICCCN*.
- [9] Tiffany Yu-Han Chen, Lenin Ravindranath, Shuo Deng, Paramvir Bahl, and Hari Balakrishnan. 2015. Glimpse: Continuous, Real-Time Object Recognition on Mobile Devices. In *Proc. ACM SenSys*. 155–168.
- [10] Xiangyu Chen, Jixiang Luo, Jingyu Xu, Fangqiu Yi, Chi Zhang, and Xuelong Li. 2025. Generative Video Compression: Towards 0.01% Compression Rate for Video Transmission. *arXiv preprint arXiv:2512.24300* (2025).
- [11] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. 2023. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. In *Proc. Robotics: Science and Systems (RSS)*.
- [12] Yuntao Dai, Hang Gu, Teng Wang, Qianyu Cheng, Yifei Zheng, Zhiyong Qiu, Lei Gong, Wenqi Lou, and Xuehai Zhou. 2025. ActionFlow: A Pipelined Action Acceleration for Vision Language Models on Edge. *arXiv preprint arXiv:2512.20276* (2025).
- [13] Kuntai Du, Ahsan Pervaiz, Xin Yuan, Aakanksha Chowdhery, Qizheng Zhang, Henry Hoffmann, and Junchen Jiang. 2020. Server-Driven Video Streaming for Deep Learning Inference. In *Proc. ACM SIGCOMM*. 557–570.
- [14] Juntao Gao, Feiyang Ye, Jing Zhang, and Wenjing Qian. 2025. Compressor-VLA: Instruction-Guided Visual Token Compression for Efficient Robotic Manipulation. *arXiv preprint arXiv:2511.18950* (2025).
- [15] Deniz Gündüz, Zhijin Qin, Inaki Estella Aguerri, Harpreet S. Dhillon, Zhaohui Yang, Aylin Yener, Kai Kit Wong, and Chan-Byoung Chae. 2023. Beyond Transmitting Bits: Context, Semantics, and Task-Oriented Communications. *IEEE Journal on Selected Areas in Communications* 41, 1 (2023), 5–41.
- [16] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. 2023. Mastering Diverse Domains through World Models. *arXiv preprint arXiv:2301.04104* (2023).
- [17] W. P. M. H. Heemels, Karl Henrik Johansson, and Paulo Tabuada. 2012. An Introduction to Event-Triggered and Self-Triggered Control. In *Proc. IEEE Conference on Decision and Control (CDC)*. 3270–3285.
- [18] Peiwen Jiang, Jiajia Guo, Chao-Kai Wen, Shi Jin, and Jun Zhang. 2025. Semantic Communications with World Models. *arXiv preprint arXiv:2510.24785* (2025).
- [19] Tobias Jülg, Khaled Gamal, Nisarga Nilavadi, Pierre Krack, Seongjin Bien, Michael Krawez, Florian Walter, and Wolfram Burgard. 2026. VLAagents: A Policy Server for Efficient VLA Inference. *arXiv preprint arXiv:2601.11250* (2026).

- [20] Hyeonjun Kim, Jegwang Ryu, Sangbeom Ha, Junhyeok Lee, Jun-Hyuk Kim, Hyemin Ahn, and Jaeho Lee. 2026. Learned Image Compression for Vision-Language-Action Models. *arXiv preprint arXiv:2606.16253* (2026).
- [21] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. 2024. OpenVLA: An Open-Source Vision-Language-Action Model. In *Proc. Conference on Robot Learning (CoRL)*.
- [22] Marios Kountouris and Nikolaos Pappas. 2021. Semantics-Empowered Communication for Networked Intelligent Systems. *IEEE Communications Magazine* 59, 6 (2021), 96–102.
- [23] Nan Li, Yansha Deng, and Dusit Niyato. 2025. Goal-Oriented Semantic Communication for Wireless Video Transmission via Generative AI. *arXiv preprint arXiv:2502.20927* (2025).
- [24] Xiangyu Li, Huaizhi Tang, Xin Ding, Weijun Wang, Ting Cao, and Yunxin Liu. 2026. OxyGen: Unified KV Cache Management for VLA Inference under Multi-Task Parallelism. *arXiv preprint arXiv:2603.14371* (2026).
- [25] Yuanqi Li, Arthi Padmanabhan, Pengzhan Zhao, Yufei Wang, Guoqing Harry Xu, and Ravi Netravali. 2020. Reducto: On-Camera Filtering for Resource-Efficient Real-Time Video Analytics. In *Proc. ACM SIGCOMM*. 359–376.
- [26] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. 2023. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In *Proc. NeurIPS Datasets and Benchmarks Track*.
- [27] Zhenyu Liu, Yi Ma, Rahim Tafazolli, and Zhi Ding. 2025. Resi-VidTok: An Efficient and Decomposed Progressive Tokenization Framework for Ultra-Low-Rate and Lightweight Video Transmission. *arXiv preprint arXiv:2510.25002* (2025).
- [28] Chong Ma, Taiyi Su, Jian Zhu, Jianjun Zhang, Zitai Huang, Yi Xu, and Hanli Wang. 2026. PiL-World: A Chunk-Wise World Model for VLA Policy-in-the-Loop Evaluation. *arXiv preprint arXiv:2606.05773* (2026).
- [29] Mohammad H. Mamduhi, Adam Molin, Domagoj Tolić, and Sandra Hirche. 2017. Error-Dependent Data Scheduling in Resource-Aware Multi-Loop Networked Control Systems. *Automatica* 81 (2017), 209–216.
- [30] Marek Miskowicz. 2006. Send-on-Delta Concept: An Event-Based Data Reporting Strategy. *Sensors* 6, 1 (2006), 49–63.
- [31] Marek Miskowicz. 2007. Asymptotic Effectiveness of the Event-Based Sampling According to the Integral Criterion. *Sensors* 7, 1 (2007), 16–37.
- [32] Adam Molin and Sandra Hirche. 2013. On the Optimality of Certainty Equivalence for Event-Triggered Control Systems. *IEEE Trans. Automat. Control* 58, 2 (2013), 470–474.
- [33] Arvind Narayanan, Eman Ramadan, Rishabh Mehta, Xinyue Hu, Qingxu Liu, Rostand A. K. Fezeu, Udhaya Kumar Dayalan, Saurabh Verma, Peiqi Ji, Tao Li, Feng Qian, and Zhi-Li Zhang. 2020. Lumos5G: Mapping and Predicting Commercial mmWave 5G Throughput. In *Proc. ACM Internet Measurement Conference (IMC)*. 176–193.
- [34] Arvind Narayanan, Xumiao Zhang, Ruiyang Zhu, Ahmad Hassan, Shuowei Jin, Xiao Zhu, Xiaoxuan Zhang, Denis Rybkin, Zhengxuan Yang, Zhuoqing Morley Mao, Feng Qian, and Zhi-Li Zhang. 2021. A Variegated Look at 5G in the Wild: Performance, Power, and QoE Implications. In *Proc. ACM SIGCOMM*. 610–625.
- [35] Vinh Hao Nguyen and Young Soo Suh. 2008. Networked Estimation with an Area-Triggered Transmission Method. *Sensors* 8, 2 (2008), 897–909.
- [36] Vinh Hao Nguyen and Young Soo Suh. 2009. Networked Estimation for Event-Based Sampling Systems with Packet Dropouts. *Sensors* 9, 4 (2009), 3078–3089.
- [37] Daojie Peng, Fulong Ma, Bingtao Wang, Sheng Wang, and Jun Ma. 2026. Latency-Tolerant Cloud-Edge Collaborative Vision-Language-Action Models via Emergent Representational Specialization. *arXiv preprint arXiv:2608.00569* (2026).
- [38] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, et al. 2025. $\pi_{0.5}$: A Vision-Language-Action Model with Open-World Generalization. *arXiv preprint arXiv:2504.16054* (2025).
- [39] Darijo Racá, Dylan Leahy, Cormac J. Sreenan, and Jason J. Quinlan. 2020. Beyond Throughput, the Next Generation: A 5G Dataset with Channel and Context Metrics. In *Proc. ACM Multimedia Systems Conference (MMSys)*. 303–308.
- [40] Young Soo Suh. 2007. Send-on-Delta Sensor Data Transmission with a Linear Predictor. *Sensors* 7, 4 (2007), 537–547.
- [41] Paulo Tabuada. 2007. Event-Triggered Real-Time Scheduling of Stabilizing Control Tasks. *IEEE Trans. Automat. Control* 52, 9 (2007), 1680–1685.
- [42] Jiaming Tang, Yufei Sun, Yilong Zhao, Shang Yang, Yujun Lin, Zhuoyang Zhang, James Hou, Yao Lu, Zhijian Liu, and Song Han. 2025. VLASH: Real-Time VLAs via Future-State-Aware Asynchronous Inference. *arXiv preprint arXiv:2512.01031* (2025).
- [43] Zuojin Tang, Shengchao Yuan, Xiaoxin Bai, Zhiyuan Jing, De Ma, Gang Pan, and Bin Liu. 2026. One Token Per Frame: Reconsidering Visual Bandwidth in World Models for VLA Policy. *arXiv preprint arXiv:2605.07931* (2026).
- [44] Lingyi Wang, Tingyu Shui, Walid Saad, and Pascal Adjakple. 2026. World Model-Enabled Causal Digital Twins for Semantic Communications in Physical AI Systems. *arXiv preprint arXiv:2605.16547* (2026).
- [45] Jialong Wu, Shaofeng Yin, Ningya Feng, Xu He, Chaochao Li, Jianye Wang, and Mingsheng Long. 2024. iVideoGPT: Interactive VideoGPTs are Scalable World Models. In *Proc. NeurIPS*.
- [46] Siyu Xu, Yunke Wang, Chenghao Xia, Dihao Zhu, Tao Huang, and Chang Xu. 2025. VLA-Cache: Efficient Vision-Language-Action Manipulation via Adaptive Token Caching. In *Proc. NeurIPS*.
- [47] Wanting Yang, Zehui Xiong, Yanli Yuan, Wenqiao Jiang, Tony Q. S. Quek, and Mérouane Debbah. 2024. Agent-Driven Generative Semantic Communication with Cross-Modality and Prediction. *arXiv preprint arXiv:2404.06997* (2024).
- [48] Hang Yin, Li Qiao, Yu Ma, Shuo Sun, Kan Li, Zhen Gao, and Dusit Niyato. 2025. Generative Video Semantic Communication via Multimodal Semantic Fusion with Large Model. *arXiv preprint arXiv:2502.13838* (2025).
- [49] Cheng Yuan, Zhening Liu, Jiashu Lv, Jiawei Shao, Yufei Jiang, Jun Zhang, and Xuelong Li. 2025. Task-Oriented Feature Compression for Multimodal Understanding via Device-Edge Co-Inference. *IEEE Transactions on Mobile Computing* (2025).
- [50] Ben Zhang, Xin Jin, Sylvia Ratnasamy, John Wawrzyniak, and Edward A. Lee. 2018. AWStream: Adaptive Wide-Area Streaming Analytics. In *Proc. ACM SIGCOMM*. 236–252.
- [51] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. 2023. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. In *Proc. Robotics: Science and Systems (RSS)*.
- [52] Tianle Zhu, Haohua Que, Handong Yao, Hongyi Xu, and Zhipeng Bao. 2026. DinoLink: A Token-Centric Representation Compression Framework for Bandwidth-Constrained Collaborative V2X Perception. *arXiv preprint arXiv:2606.26398* (2026).